

NETFLIX DATA ANALYSIS USING POWER BI

1. Introduction

- This project analyzes Netflix content and customer viewing behavior using **Power BI**.
- The analysis covers customer demographics, content performance, ratings, genres, subscriptions and watch time.
- The project converts raw Netflix datasets into meaningful business insights using interactive dashboards.

2. Problem Statement

- Netflix has large amounts of content and customer data, making it difficult to identify useful patterns manually.
- The project aims to identify popular content, customer preferences, subscription performance and engagement trends.
- The analysis helps support data-driven content and customer strategy decisions.

3. Objectives

- To clean and analyze Netflix content and customer datasets.
- To identify trends in **content popularity, customer ratings, watch time and subscriptions**.
- To create an interactive Power BI dashboard and provide business recommendations.

4. Dataset Description

- **Netflix Shows Dataset:** Contains show ID, title, type, genre, country, director, cast, release year and rating.
- **Customer Ratings Dataset:** Contains customer ID, age, gender, country, subscription type, watch time and rating.
- The two datasets are connected using **show_id**.

5. Tools Used

- **Excel:** Data cleaning and preprocessing.
- **Power BI:** Data visualization and interactive dashboard creation.
- **DAX:** KPI calculations and analytical measures.

6. Task Details

Task 1 – Data Cleaning

- Checked missing values and duplicate records.

- Standardized dates, ratings and data types.

Task 2 – Customer Demographics

- Customers were grouped into different age categories.
- The **60+ age group** was the largest segment.

Task 3 – Subscription vs Watch Time

- Compared Basic, Standard and Premium plans.
- **Standard** had the highest customer count and total watch time.

Task 4 – Top 10 Content

- Identified the most-watched movies and TV shows.
- Compared content based on total watch time.

Task 5 – Genre by Country

- Analyzed popular genres across different countries.
- Used country-level visualization to identify regional preferences.

Task 6 – Customer Ratings

- Analyzed ratings from **1 to 5**.
- Compared average ratings by genre and country.

Task 7 – Trending Content

- Compared **New Content vs Older Content**.
- New content had a slightly higher average rating in the sample.

Task 8 – Subscription Market Share

- Compared customer distribution across subscription plans.
- Identified the dominant subscription type.

Task 9 – Customer Retention

- Planned to compare single-show and multiple-show customers.
- Actual retention analysis was limited because each customer has one recorded show interaction.

Task 10 – High vs Low Rated Content

- Compared ratings across genres.
- Compared Movies and TV Shows based on average ratings.

Task 11 – Director & Cast Analysis

- Analyzed the relationship between directors, cast and ratings.
- Identified highly rated directors based on available data.

Task 12 – Content Localization

- Compared engagement and ratings across countries.
- Identified opportunities for localized content strategies.

Task 13 – Release Timing

- Analyzed content additions by month.
- Compared monthly content volume with customer ratings.

Task 14 – Customer Engagement Dashboard

- Created visuals for watch time, ratings, genres and subscriptions.
- Added slicers for **country, age group and subscription type**.

Task 15 – Competitor Benchmarking

- Compared Netflix performance conceptually across countries.
- Actual competitor comparison was not possible because competitor data was unavailable.

7. Results / Key Findings

- The **60+ age group** was the largest customer segment.
- **Standard subscription** showed the highest customer count and total recorded watch time.
- New content showed slightly higher average ratings than older content.
- Genre and country analysis revealed differences in customer preferences.
- Power BI provided an interactive view of customer and content performance.

8. Business Recommendations

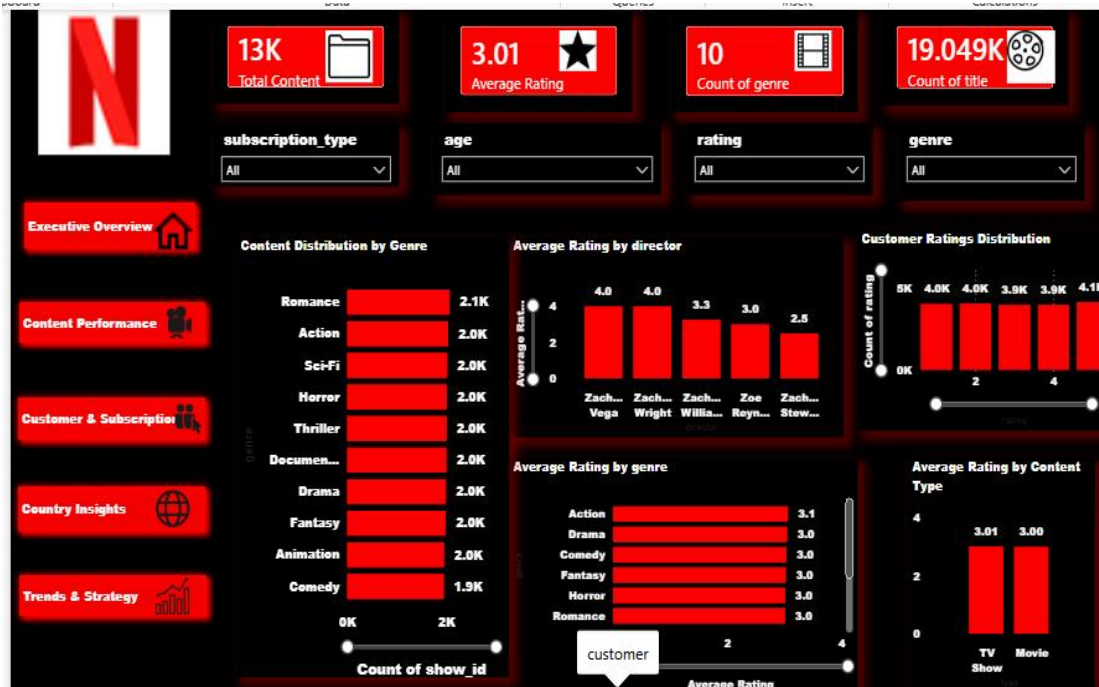
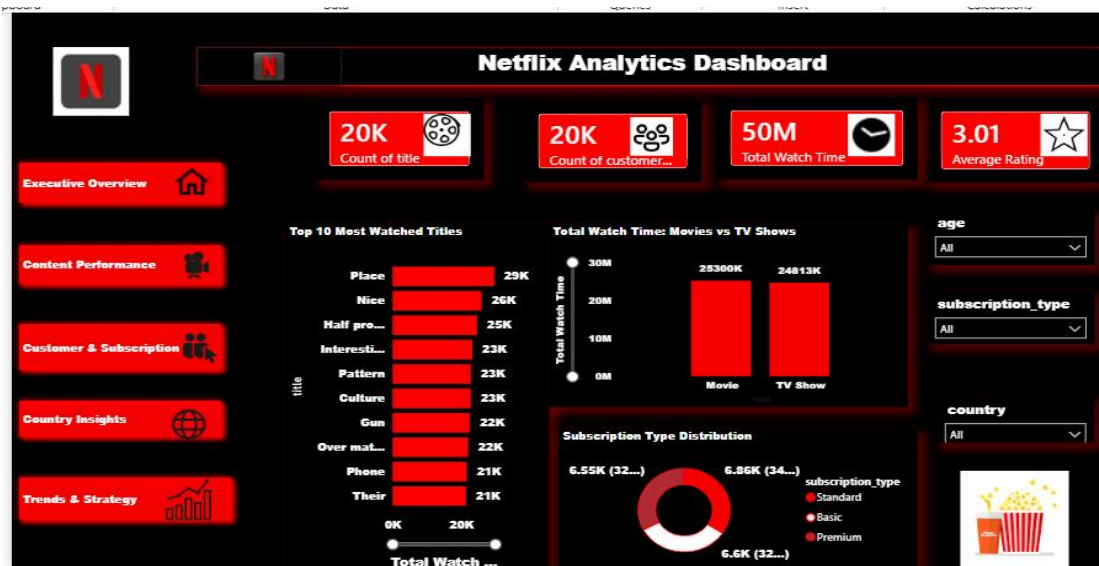
- Focus on highly watched and highly rated genres.
- Develop **country-specific content strategies** based on regional preferences.
- Monitor subscription engagement and improve customer retention tracking.
- Use monthly content trends to support future content release planning.

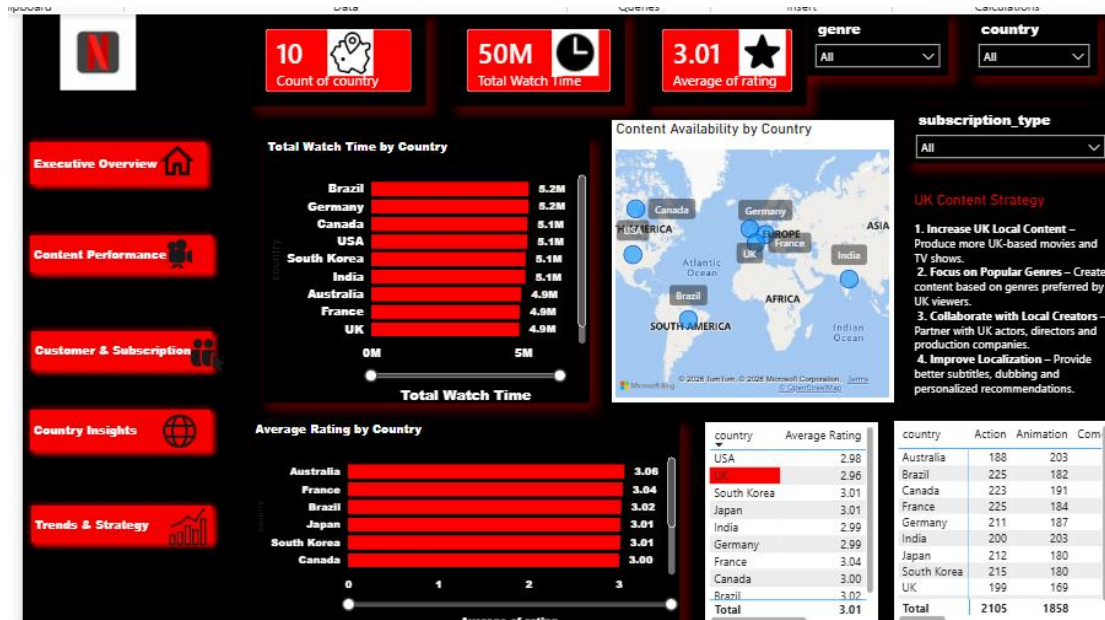
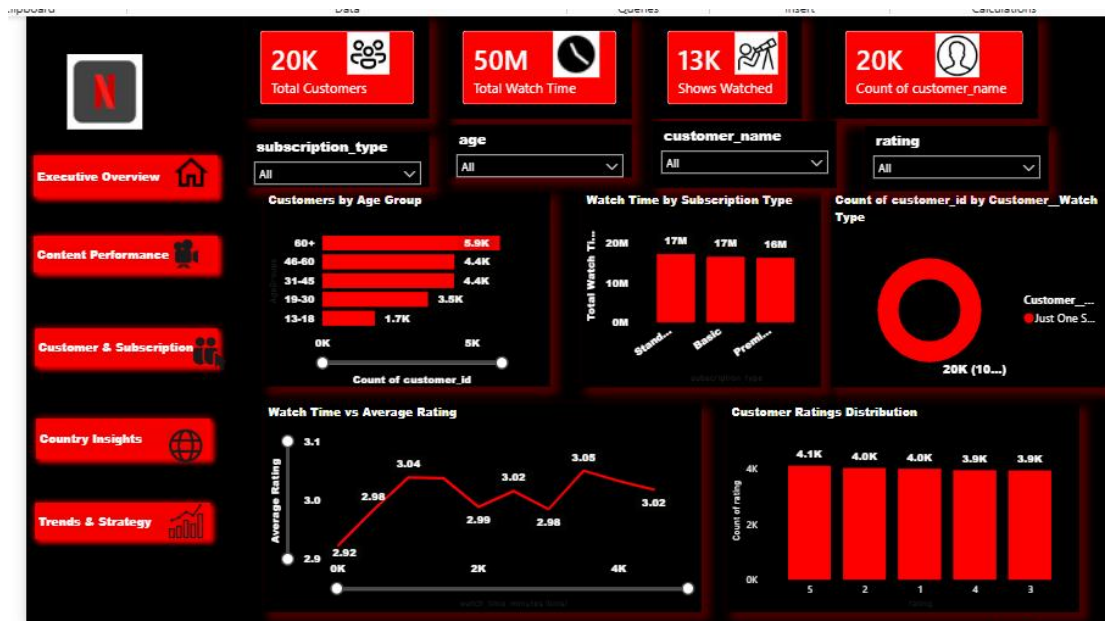
9. Conclusion

- The project successfully analyzed Netflix content and customer data using **Excel and Power BI**.
- The dashboard provides useful insights into **customers, subscriptions, content, genres, ratings and countries**.

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- The screenshot displays a data visualization interface with two data sources and a relationship line. The first data source, 'netflix_dataset_200...', lists fields: 'cast', 'Content Age', 'country', 'date_added', 'director', 'duration', 'genre', 'Month', and 'Month Number'. The second data source, 'netflix_customer_rating...', lists fields: 'age', 'AgeGroups', 'country', 'Customer_Watch Type', 'customer_id', 'customer_name', 'gender', 'rating', and 'show_id'. A relationship line connects the 'country' field of the first source to the 'country' field of the second source. The line has a '1' on the first source side and an '*' on the second source side, indicating a one-to-many relationship.







Results

- Standard subscription recorded the highest customer count and total watch time in the dataset.
- The 60+ age group was the largest customer segment, showing strong engagement in the sample.
- Action was among the highest-rated genres, while new content received slightly higher average ratings than older content.
- The Power BI dashboard identified content popularity, customer engagement, subscription trends, ratings and country-wise preferences.
- The analysis provides useful insights for content planning, regional strategies and customer engagement improvement.